

Wearables in real life: A qualitative study of experiences of people with epilepsy who use home seizure monitoring devices

Lotte Schelde Olsen^{a,b,*}, Jonas Munch Nielsen^{a,b}, Charlotte Simonjy^{c,d,e}, Troels Wesenberg Kjær^{a,b}, Malene Beck^{a,c}

^a Department of Neurology, Zealand University Hospital, Region Sjælland, Vestermærksvej 11, 4000 Roskilde, Denmark

^b Department of Clinical Medicine, University of Copenhagen, Blegdamsvej 3B, 2200 Copenhagen N, Denmark

^c Institute of the Regional Health University of Southern Denmark, 5230 Odense, Denmark

^d The Research Unit PROgrez, Department of Physiotherapy and Occupational Therapy, Næstved-Slagelse-Ringsted Hospitals, Region Zealand, 4200 Slagelse, Denmark

^e Department of Research Naestved, Slagelse and Ringsted Hospitals, Denmark

ARTICLE INFO

Article history:

Received 11 June 2021

Revised 18 October 2021

Accepted 21 October 2021

Available online 13 November 2021

Keywords:

Epilepsy

Wearables

Patient perspectives

Seizure detection

Qualitative research

Patient experiences

ABSTRACT

Objective: To explore the experiences of people with epilepsy using wearables for home seizure monitoring.

Methods: Nine people with epilepsy participated in eighteen semistructured individual interviews before and after home monitoring with wearable seizure monitoring equipment. An open-ended interview guide was used to encourage the participants to elaborate on their thoughts and experiences. Interviews were analyzed using a three-level process inspired by the philosopher Max van Manen.

Results: The overall findings illustrate that patients experienced *being placed in the spotlight* when wearing wearables. The meaning of being in this spotlight is reflected in three themes: *Becoming vulnerable through exposure*, *Standing alone while being with others*, and *Having a renewed life situation*. The analysis and interpretation showed that although the participants expressed readiness to use the wearables, they were less willing to do so after a few days of monitoring. The visibility of the devices influenced how they experienced themselves and were perceived by others.

Conclusion: For people with epilepsy, wearables are more than just technical tools; they have a significant existential impact on everyday life. Wearables spotlight the epilepsy condition, and this causes people with epilepsy to experience an existential disruption, as they experience being exposed and vulnerable. This results in a renewed way of perceiving oneself. Nevertheless, wearables also validate epilepsy symptoms, thereby reducing the uncertainty related to epilepsy.

© 2021 The Author(s). Published by Elsevier Inc. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

1. Introduction

In the past decade, numerous wearable health devices that can be used to monitor sleep, physical activity, and health have emerged [1,2], and this technology is increasingly utilized in the healthcare sector [1,3]. Wearable health devices to monitor epilepsy (hereinafter wearables) are intended to function as a solution for objective seizure quantification and alarming [4]. Subjective seizure counting has long been regarded as unreliable [5], but it is still clinically relied upon due to the lack of a good objective alternative [6]. Despite medical treatment, up to 30% of people with epilepsy (PWE) are not seizure free [7]. The unpredictable nature of seizures can cause distress and lead to social isolation

[8]. Wearables that provide reliable seizure monitoring can individualize and improve health and self-care, allowing for more personalized health care [2].

When monitored in an epilepsy monitoring unit (EMU), PWE are exposed to specialized medical equipment, such as video monitoring, that can compromise personal freedom and privacy [9]. EMU monitoring serves an important diagnostic role, but it is infeasible and excessive for long-term outpatient seizure monitoring. New methods of seizure detection are being developed for this patient group, and in recent years, an increasing number of wearable devices and digital technologies have become available [10]. Studies have shown that 80–94% of PWE are willing to explore the use of wearables [10,11]. The visual appearance, comfort, and accessibility of such devices greatly affect how willing PWE are to use wearables outside the hospital. Nonetheless, the fear of social exclusion and stigma can hinder the use of wearables [12].

* Corresponding author at: Vestermærksvej 11, 4000 Roskilde, Denmark.

E-mail address: losol@regionsjaelland.dk (L.S. Olsen).

Previous research on the readiness for wearables among PWE [13] has shown that wearables are valuable because they enable a better understanding of the life situation of PWE. The technology is thus more than merely a technical tool: it can be considered an esthetic and existential artifact that acts as a lifeline between PWE and their healthcare providers. Wearables legitimize the signs and symptoms of epilepsy, perhaps leading to improved acknowledgment from health professionals, friends, and relatives. Indeed, wearables have been identified as a tool that could potentially improve the quality of life among PWE [13].

Studies have investigated the potential barriers and facilitators related to the use of wearables among PWE. Most research has focused on the technical and visual aspects of devices, and little is known about the potential impact of wearables on the everyday lives of PWE. There is sparse knowledge of the experiences of PWE in regard to wearables, especially during home monitoring (i.e., “in-the-wild” monitoring). In the wild indicates the use of wearables in an everyday setting, such as during work, shopping, and sleep. The aim of this study was to explore the experiences of PWE regarding in-the-wild use of wearables.

2. Method

2.1. Design

A qualitative hermeneutic phenomenological approach inspired by the philosopher Max van Manen was used [14]. With this approach, it is possible to gain new knowledge of how it is to be a person using wearables in the wild. The methodology combines a descriptive presentation of lived experiences with an interpretive understanding and is used in this work to explore what is significant to PWEs. A lived experience description is a vivid textual account of an experience just as it was lived through [14]. It describes a particular instance of an experience in concrete terms, avoiding abstraction, introductions, rationalizations, causal explanations, generalizations, and interpretations [14]. Thus, the goal of this study was to disclose *what* was experienced and *how* it was experienced as a means of examining and better understanding new meanings and appreciations related to the use of wearables.

2.2. Participants and settings

Participants were invited to participate in the study when admitted to the EMU or when visiting the outpatient epilepsy clinic at a regional university hospital in Denmark. These participants were approached by one member of the clinical team at the department and subsequently by the researcher JMN. Ten participants were invited to participate; one declined the invitation due to personal reasons. Five women and four men were included

in the study (Table 1). Participants ranged in age from 28 to 44 years. Eight participants had had epilepsy for more than six years, and one participant was diagnosed within nine months prior to inclusion in the study. The authors recruited through a purposive sampling process [15] and strove to recruit participants of both genders and of varied ages and disease durations. This participant diversity provided varied contexts within which to investigate the phenomenon of using wearables in the wild. The participants were required to speak and understand Danish fluently. Furthermore, participants who were not considered able to manage the wearables (e.g., due to cognitive decline, psychiatric comorbidities or abuse problems) were excluded from the study. All participants were informed verbally and in writing about the purpose of the study and gave informed consent.

In the-wild monitoring was offered only for research purposes. However, most participants were very interested in participating because they hoped that this monitoring would improve their own and their healthcare professionals' knowledge of epilepsy.

2.3. Wearables

Three wearable devices were used to monitor each participant. The devices included a portable electroencephalography amplifier with two channels, an electrocardiography device fixed to the participant's sternum with a patch electrode, and one to four accelerometers attached to the limbs using small patches. Each participant was instructed by LSO and JMN on how to manage the wearables and was thoroughly prepared for possible technical challenges associated with using the devices. This process took approximately an hour. Participants were encouraged to wear the devices day and night for up to five consecutive days. However, they could freely choose when they wanted to wear the equipment, as it could easily be removed and later reattached. Furthermore, the participants were encouraged to keep seizure diaries and to take notes regarding technical difficulties, adverse events, and notable experiences associated with using the equipment. While being monitored, the participants were encouraged to follow their regular routines, including going to work and socializing; this was possible due to the wearable nature of the devices and the lack of video monitoring. The participants were given the number of a direct telephone hotline to the investigators to ensure that they felt safe and confident when managing the wearables. One participant who experienced only nocturnal seizures chose to use the wearables only in the evening and at night. No technical problems or adverse events were reported during in-the-wild monitoring.

2.4. Data collection

In this study, data were collected through interviews conducted in the homes of the PWEs. In these interviews, the researchers

Table 1
Age, sex, disease, duration, and monitoring time of the study participants.

Participant	Age (yrs)	Sex	Marital status	Employment	Epilepsy duration	Seizure type	Seizure frequency	Monitoring time (days)
P1	44	F	Living with partner	Employed	24 years	FIAS	1/month	3
P2	34	F	Living with partner	Unemployed	9 months	Unknown	Daily	4
P3	42	F	Living with partner	Employed	7 years	FTBTCS	1–2/week	3
P4	22	M	Living alone	Unemployed	6 years	FTBTCS	1/month	4*
P5	42	F	Living with partner	Employed	7 years	FIAS	1/week	4
P6	29	M	Living alone	Unemployed	11 years	FTBTCS	Daily	3
P7	44	F	Living with partner	Unemployed	24 years	FIAS	1/year	4
P8	30	M	Living alone	Employed	< 20 years	FIAS	2/month	3
P9	28	M	Living alone	Student	10 years	NFFL	Daily	4

FIAS: Focal impaired awareness seizures.

FTBTCS: Focal temporal lobe epilepsy with secondary generalizations.

NFFL: Nocturnal focal frontal lobe epilepsy.

focused on discovering patterns of meanings and variations that described the essential meaning of what it is like to be a person using wearables in the wild [14]. Hermeneutic phenomenology is the study of the lifeworld or of the world as experienced by an individual. This can be explored through the five lifeworld existentials: lived body, lived self-other, lived space, lived time, and lived things and technology [14]. These existentials are fundamental, common themes that lie within all human lives regardless of the individuals' histories, social situations, and cultures. According to van Manen, existentials are helpful as a guide for reflections through the research process [16]. Thus, the five lifeworld existentials were used to structure the interviews, with the purpose of exploring all aspects of the participants' experiences of the phenomenon of using wearables. Although existentials will always overlap with each other, in a given situation one existential might be more pervasive than another [16]. In this study, the existentials of the lived body and lived self-other naturally came more into focus, resulting in more questions being directed to these existentials. The questions in the interview guide were not developed with reference to the known epilepsy literature, as the intention was to maintain an open attitude throughout the data collection.

In practice, data were collected by inviting PWE to participate in two interviews. The interviews were conducted in the participants' homes to provide a visual context for the use of wearables in the wild and used a semistructured interview guide (Table 2) inspired by Kvale [17]. With the hermeneutic phenomenological approach, the interviewer strived to create an open atmosphere by listening and paying attention to what was said. The interview guide was used to maintain a focus on the phenomenon while also responding to the participant's described experiences and thoughts.

A total of eighteen interviews were conducted. The initial interview was conducted before monitoring began to gain insight into any immediate thoughts the participant might have on using the wearable in the wild. After monitoring, a comprehensive interview was conducted to elaborate on experiences the participant had after wearing the devices in the wild. Interviews were conducted until data saturation occurred [18].

All interviews were conducted by LSO as individual face-to-face conversations. The interviewer had no clinical experience with epilepsy, a fact that provided independence and an open attitude during the interviews and subsequent interpretation [19]. The participants were encouraged to provide responses regarding their experiences and expectations that were as thorough and descriptive as possible and were also encouraged to share examples from their everyday lives related to the wearables.

Table 2
Examples of questions from the interview guide.

Intro	<i>What are you occupied with at the moment? In relation to the wearables – how do you describe the days you have been using them?</i>
Lived body	<i>Can you please try to explain to me how you experience wearing the wearables on your body?</i>
Lived self-other	<i>If I should try to imagine being you, how would you describe what is important to you when you attend social gatherings wearing these wearables?</i>
Lived space	<i>What is it like to be you and wear these wearables when you are outside your home? Try to think of a specific situation where you had to leave home – how did you experience this?</i>
Lived things and technology	<i>Which opportunities can these wearables provide you now and potentially in the future in relation to your epilepsy?</i>
Lived time	<i>Can you think of a particular moment through the day when you come to think of the wearables?</i>
Conclusion	<i>How was your overall experience of using these wearables? Do you have anything you would like to add?</i>

Data were collected between July 2020 and October 2020. As the data were collected during the COVID-19 pandemic, the researchers were attentive to how this might affect the participants' opportunities to perform daily activities and to be physically present at their workplaces. However, during this period, pastime activities and workplaces were not affected by a lockdown in Denmark. All interviews were audio-recorded and transcribed verbatim, yielding 101 pages of text. The excerpts used in this article were translated into English by LSO, with JMN and MB confirming translation accuracy or suggesting improvements until consensus was reached.

2.5. Ethical considerations

The ethical guidelines set forth in the Helsinki Declaration and the guidelines offered by the Nordic Nurses' Federation were followed during the study and data collection. The study was approved by the Regional Committee on Health Research Ethics (SJ-725) and by the regional data protection agency (REG-121-2018). When interviewing participants, the interviewer must be aware of the asymmetric relationship between the interviewer and the participant [17]. Humbleness and attentiveness were strived for during the interviews to avoid causing the participants any harm when they were interviewed about personal themes [20].

2.6. Data analysis

To gain insight into the experience of using wearables in the wild, the interviews were interpreted by performing a thematic analysis that was inspired by the analytic steps (Fig. 1) described by van Manen [14]. This method included dynamic switching between wholistic, selective, and detailed phases, opening up insight into the existential phenomenon [14]. In practice, this meant that the interviews were first read repeatedly and reflected upon in their entirety (wholistic phase). The overall research question “how do people with epilepsy experience using wearables during in-the-wild monitoring?” guided the search for wholistic descriptions and more general descriptions of the phenomenon. This made it possible to get a sense of “what was going on” and “what has been said” during the interviews. Second, in the selective phase, quotes were identified that had special significance with respect to how the participants experienced wearing the wearables. To identify quotes that were endowed with significant meaning, we created themes that enabled us to capture meaning and to better reflect on what was said during the interviews. This phase was characterized as a process in which the choice and selection of themes were discussed thoroughly and continuously within the research group to validate the quality and trustworthiness of the themes. Finally, in the detailed phase, individual sentences and paragraphs were examined with the purpose of choosing representative and illustrative quotes that exemplified the meaning and interpretation within the theme and subthemes in the analysis [14]. In practice, this phase leads to better insight into PWE's experiences of using wearables in the wild. Author LSO performed the initial analysis, and all authors then discussed the presented themes and findings as a basis for validating their meaning and interpretation.

3. Results

3.1. The essence: being placed in the spotlight

The essential meaning of using wearables in the wild to people living their everyday lives with epilepsy is characterized by the

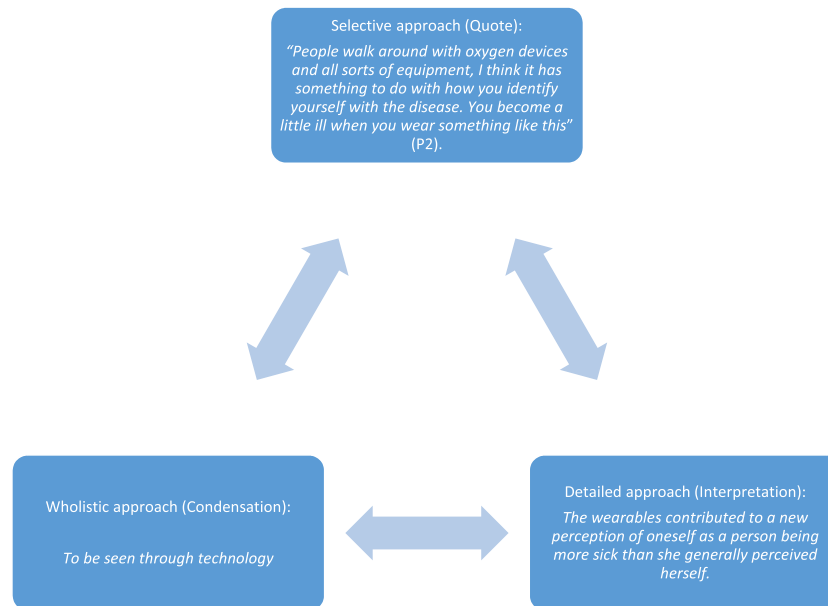


Fig. 1. Examples from the analytical process.

experience of being placed in the spotlight. The participants experienced wearables as a constant reminder of their epilepsy that visually confronted both themselves and people around them with the fact that their lives were attuned by the condition. The participants shared that this experience made them feel affected by their condition in an existential way; hence, wearables encouraged new insights into their being-in-the-world. At the same time, wearables provide a long-awaited experience of an attempt to validate their epilepsy symptoms. This reduced uncertainty caused by the epilepsy provides comfort and hope for clarification of their experienced symptoms. Furthermore, the analysis showed that although the participants initially expressed readiness to use the wearables, they were less willing to do so after a few days of monitoring. The visibility of the devices influenced how they experienced themselves and how they were perceived by others.

The essential meaning is further elaborated in three themes: 3.1.1 Becoming vulnerable through exposure; 3.1.2. Standing alone while being with others; 3.1.3. Having a renewed life situation.

3.1.1. Becoming vulnerable through exposure

Participants shared their feeling that wearables served as a reminder that something was different about them. Furthermore, they disclosed that wearing the devices made their condition and symptoms more readily visible, both when they were alone and when they were with others. This visual aspect of the wearables, which included the portable electroencephalography amplifier with visible electrodes, electrocardiography, and accelerometers, brought them into the spotlight, and this made them feel uncomfortable and vulnerable. The participants indicated that they found this unpleasant because being in this type of spotlight was experienced as exposing epilepsy. Furthermore, it influenced how they dealt with the disease. For the participants, every single piece of the wearables represented a bodily exposure of them as vulnerable patients with epilepsy. Because the devices shed light on their condition, they considered themselves more fragile than others, and wearing the devices reminded them of how affected they had been and of the inescapable nature of their medical condition. Some had learned to live ordinary everyday lives in which they were rarely confronted with epilepsy. However, the visibility of the wearables focused their attention on the experience of their condition, including its potentially life-threatening nature. This confrontation

resulted in uncertainty and thoughts about future aspects of living with epilepsy. The participants revealed that the wearables prompted peripheral and unknown relations to ask questions about their epilepsy to which they had to respond. The way participants felt about having to discuss their epilepsy with others was related to how they identified themselves as persons. One participant said:

"(...) It is nice that we take care of each other. But if you are not the type of person who wants to stand out from the crowd, and are in a position where you do not know exactly what you should reply to questions, then it can be a bit uncomfortable, because I don't like answering and I don't like calling attention to myself." [P2]

The visibility of the wearables prevented participants from hiding their epilepsy from their friends and families. This created a vulnerability about their life situations that was influenced by the people around them. The participants felt that the increased attention was an invitation to openness related to their epilepsy. Most experienced a negative attitude toward this attention. According to the participants, in an ordinary life with epilepsy, they were able to decide for themselves whom they wished to tell about their condition, except in situations in which they had seizures in public. However, using wearables required them to tell other people about it. This unwanted focus made the participants uncertain about how other people viewed them and caused them to fear that others would consider them vulnerable; being unable to decide for themselves with whom they would share their condition resulted in the experience of wearables being perceived as uncontrollable elements. This could trigger uncertainty, as the participants did not know when they would be confronted with questions about their epilepsy. In that sense, the participants revealed that they spent much energy trying to hide the wearables. One participant said:

"(...) yesterday, I spent time with my study group and I had to take the bus. I did not think it was fun to wear it in that situation, so I had a big sweater on, trying to prevent people from looking too much at me (...)" [P1]

Hiding wearables allowed participants to avoid unwanted attention from other people. This also served to reduce the constant reminder that their lives were affected by epilepsy. According

to the participants, they were attentive to moments and situations in which they wished to hide the wearables. During the analysis, it became clear that using wearables in public served to make the invisible disease visible. The visibility caused uncertainty for the patients, and attempts to hide the wearables might fail. This meant that the participants experienced a lack of control over who would know about their condition. Participants shared that this had a major impact on how they related to their epilepsy. Thus, it was important for the participants to be in control of when they wanted to inform others about their condition. A participant described this by saying:

"(...) nobody at my workplace knows that I have epilepsy, so if I had to wear this at work, I would have to explain a lot and personally it would push my limits. Everybody knew about it at my former workplace, so in that situation it would not have been that big a deal." [P5]

According to the participants, the wearables signified increased severity of their epilepsy to their family and friends. Epilepsy was not something that was spoken about in daily conversations within the family prior to utilizing the wearables. Therefore, family and friends did not always identify the participants as persons living with epilepsy. When they saw the participant using the wearables, it appeared as though the epilepsy had suddenly worsened. This could be frightening, especially for children. Being looked at differently caused the participants to experience a need to protect their loved ones, as they were aware that they were now perceived as more fragile. One participant described how her children reacted during the monitoring:

"... I could sense that they thought it was, I don't know if they were worried, but it was definitely like it became very obvious to them." [P2]

Another participant recounted the reaction she had from her daughter:

"When she realized that I had these things on, she became very scared and became convinced that now it was all very serious." [P5]

In this direct way, the relatives were impacted by transforming the participants' readiness to use wearables into an unwillingness.

3.1.2. Standing alone while being with others

The presence of children had a negative influence on the participants' perceptions of the acceptance of wearables. This affected their willingness to wear them in a way that allowed others to see them. One participant was asked to describe how using wearables among family and friends affected her everyday life:

"It (the epilepsy) is nothing we normally talk about; they assume that everything is fine. So when I suddenly wear these devices, they probably worry that now there is something wrong, and that will require a lot of explaining." [P5]

The close families and friends of the participants were attentive to the visibility of the epilepsy. One participant was told by her husband to "cover up" [P7] before leaving the house because he worried about the unwanted attention the wearables could draw. Adolescent relatives of the participants felt "embarrassed" [P2] if they were seen in public with a participant using the wearables. This negatively influenced the participants' willingness to use them. When relatives expressed these concerns, the participants became worried that the wearables would create a distinction in how other people perceived them. This caused them to experience being left out of the community and left them in a lonely position. The visual appearance of the wearables was important: the partic-

ipants shared their feeling that if the devices had been modern and small, they would not have caused the wearer to stand out from the crowd. The participants indicated that the less visible the wearables were to them and to those in their surroundings, the more willing they were to wear them. This surprised many participants because during the preliminary interviews most of the participants described themselves as indifferent to what other people thought of the wearables. They discovered that, in reality, the visual aspect had a significant impact on how they experienced everyday life and on how they were perceived by others. One participant described this, saying:

"I realized that how people think of me actually means more to me than I thought. Normally, I think that people can look all they want, but... when it came to it, I didn't think it was that pleasant." [P1]

The participants offered many suggestions about how the devices could be made more attractive. One participant said:

"(...) there exists all these modern purses like those I have over there (points at her handbags and purses in the room). If you could provide some of those (...), then it looks more modern. Perhaps the men would not want to wear them, but as a lady, then you could wear it around the belly, if that is what you want, and that would be a lot fancier." [P7]

The participants also suggested that the wearables could be made to look like smartphones or that the electrodes could be wireless. The wearables needed to reflect recognizability to maintain consistency with everyday life without drawing increased attention to themselves. The participants clearly preferred not to stand out. An example was one participant who said:

"(...) If you could make it to the size of a smartphone and you could just put it in your belt, then it wouldn't be that big of a deal, so [it would be like] walking around with a fanny pack." [P6]

3.1.3. Having a renewed life situation

Through in-the-wild monitoring, participants began to think differently about how they experienced themselves and their epilepsy. This meant that existential thoughts of how they perceived their identity in relation to being a human whose life was affected by epilepsy emerged. Although the participants had previously stated that they would be willing to wear wearables in public, they came to a new recognition of how the devices, in reality, affected their lives. This new recognition meant that the participants began to perceive wearables not simply as technical devices but as components that were rooted deeply in their existential way of being-in-the-world. Through the use of wearables, they experienced increased personal attention to their epilepsy. For participants who only rarely experienced seizures, the wearables became a clear reminder of something they usually could forget. An example of this was when a participant said:

"Well, it reminds me that the disease is present, and the seizures are possible. When I am wearing a fanny pack with all these wires, I can't really ignore it the same way as I can [normally] forget it when I'm cooking or spending time with the children through an ordinary day." [P1]

Our analysis revealed that wearables mobilized a variety of existential thoughts about living with a disease, many of which impacted everyday life. Using wearables caused the participants to feel that the disease was more in the forefront of their lives than it had been previously. This meant that they considered other aspects of living life with a disease more than they did before using wearables in the wild. An example of this was a participant who said:

(...) people walk around with oxygen devices and all sorts of equipment, I think it has something to do with how you identify yourself with the disease. You become a little sick when you wear something like this.” [P2]

For those who were newly diagnosed or had numerous seizures, wearables were seen as a valuable tool that provided them with insight into their epilepsy. Some participants had been struggling for many years to understand and legitimize their bodily signs and experiences of seizures as they relate to themselves, their surroundings, and their healthcare providers. In this situation, wearables were perceived as an acknowledgment of their experience of having epilepsy. Wearing the devices helped them reduce their uncertainty about their symptoms and condition. One participant described it this way:

“(...) to have this equipment on, that is very serious, I feel that I'm being taken seriously (...) it seems more purposeful on the part of the department [Department of Neurology], when one is examined in that way.” [P2]

Most of the participants experienced regaining control regarding their epilepsy symptoms when using wearables. They all had hopes that wearables could provide insight into seizure frequency, seizure patterns, and the localization of their epilepsy, all of which could potentially assist health care providers in improving their medical treatment. An example of this was a participant who said: *“(...) if it can detect it (the seizures), and they can say that I need another treatment, that would be great.” [P1]*

While experiencing hopefulness about a potential clarification of their symptoms, the participants also felt stressed during monitoring. Most participants had previous experience with inpatient monitoring in the EMU. In-the-wild monitoring allowed the participants to maintain everyday life with family and friends, regular activities, and work. However, even though the wearables provided an opportunity for the participants to stay in their home environment, they still experienced stress regarding monitoring the seizures. On the one hand, they were interested in recording a seizure. On the other hand, a seizure was normally something they tried to prevent. One participant said:

“I have thought a lot about it, and it is like when you work in emergency preparedness (...), it is good for the community if nothing happens, but it is really boring to be at work. And it is the same with this because I had hoped that it could not register anything. After all, I have not had a seizure while monitoring, but at the same time, I would have wanted there to be something, because it could help me localize something (...).” [P1]

The participants shared that they were equally stressed about having a seizure when in the EMU and during in-the-wild monitoring, but said that at home they were better able to relax and feel comfortable. This ambiguous situation increased attention to wearables. However, using in-the-wild monitoring caused the participants to feel that they were confronted with their epilepsy in places in which they had previously felt safe and calm. This meant that the wearables became an intruder within the experience of homeliness that PWE used to feel because they elicited new thoughts on how it is to be a person living life with epilepsy symptoms.

4. Discussion

The use of wearables by PWE placed a spotlight on their condition and caused an existential disruption in their everyday lives. We demonstrate that for PWE, wearable monitoring equipment is more than technical monitoring equipment; it affects the way in which they understand themselves as individuals with epilepsy.

Our study shows that, for PWE, wearables influence self-perception and experiences of everyday life. A previous study found that 94% of PWE were ready to use wearables for automatic seizure detection but that only 15% were willing to wear patch electrodes on the head [11]. This is consistent with the results of our previous study, in which PWE indicated a readiness to use wearables [13]. However, in this study, we provide new knowledge on how PWE's willingness to use these devices is challenged once the wearables are used. In fact, these technical tools put the life situation of PWE in the spotlight, and this causes them to experience vulnerability and embarrassment but also renewal. Thus, unforeseen barriers challenge the use of wearables in everyday life.

Visible electrodes on the head were a particular barrier because they made an otherwise invisible disease visible to others. This may explain why previous research shows that only 15% of PWE are ready to wear electrodes on the head [10]. This is supported by previous findings that the visual appearance of wearables must be nonstigmatizing and nonintrusive [12].

Our study reveals that wearables impact self-image. Thus, in our study, wearables are found to be a constant reminder to PWE of their diagnosis in an existential sense. The philosopher Drew Leder emphasized the notion of how the body becomes an object to the mind when some of its functions are at stake [21]. Leder argues that the usual disappearing body becomes “dys-appearing” when illness changes one's abilities. The consciousness becomes directed toward the parts or functions of the body that do not respond as one wants them to. Hence, these parts become more visible than they previously were [21]. Our study shows how wearables can be seen as technical artifacts that render the body dys-appearing to the people who wear them, meaning that PWE become more attentive to their bodies as abnormal and different when wearables are used. Under these circumstances, the PWE experienced their existence in a new way, caused a renewed and more visible relation to epilepsy. This implies that wearables might impact the way PWE perceive themselves and how others view them. In light of being confronted with this new way of perceiving themselves and their epilepsy, wearables make their bodies reappear. This new insight provides valuable knowledge to health professionals who work in epilepsy monitoring when considering the practice of health pedagogy. Based on our findings, we argue that it is important that health professionals also consider the existential impacts similar technical devices can have on people who are being monitored.

Our study illustrates that for PWE wearables can be a tool that allow them to experience acknowledgment of their symptoms. Thus, wearables have the potential to provide an experience of legitimization in relation to everyday life with epilepsy. This is consistent with previous research [13]. Our study further elaborates on how the use of wearables can help PWE deal with uncertainty related to epilepsy symptoms. Uncertainty, here, can be defined as the inability to determine the meaning of illness-related events, a cognitive state that occurs when individuals cannot adequately categorize or structure an illness event due to insufficient cues [22]. Uncertainty in illness often occurs in situations that are unpredictable, complex, or ambiguous. Inconsistent information or lack of information also creates uncertainty. Our study reveals that living with epilepsy can mitigate and exacerbate a sense of uncertainty [22]. There is a complex relationship between the use of wearables and the experience of uncertainty in epilepsy that depends on the context and the person. The uncertainty can be decreased by the use of wearables because they provide PWE with hopeful expectations, as the improved categorization of symptoms as epileptic or nonepileptic that the wearables provide can pave the way for a better understanding of such signs.

Our study provides valuable knowledge regarding the importance of the visual aspect of wearables to PWE, especially as it is

intertwined with perceptions of uncertainty experienced by friends and relatives. Our study showed that PWE experience the visual aspect of wearables as impacting their presence within the context of family and friends. PWE experience that their relatives perceive wearables as a confronting element of epilepsy, leading to uncertainty and worry. These experiences affect the willingness of PWE to use wearables, and it came as a surprise to them that they were affected in this way. We argue that this new insight should be recognized by healthcare professionals in clinical practice so that they may prepare future PWE to expect more attention to their epilepsy than they might anticipate prior to in-the-wild monitoring. Furthermore, comfort and support should be offered by healthcare professionals during monitoring to mitigate the potential existential struggles that PWE using wearables may experience. Mishel and Clayton point out that when relatives experience uncertainty, it can impair their ability to provide support for a person [22]. Therefore, providing information and encouraging the involvement of relatives when using wearables is essential for increasing the willingness of PWE to use future health technology.

5. Strengths and limitations

The current study provides valuable insights into the perspectives of PWE who use wearables for in-the-wild seizure detection.

The participants were interviewed before and after monitoring, allowing the study to analyze both their expectations and their experiences. The participants were monitored for up to five consecutive days while they used a specific technological setup. This is a limitation, as the results may not be generalizable to longer periods of monitoring or to experiences involving the use of different equipment. The participants were free to choose when they wanted to use the wearables during the day. This allowed the study to explore the situations and circumstances under which they were willing to use the devices.

In this study, we interviewed nine participants. The sample size is not extremely relevant in qualitative research since the number of participants does not reflect the quality of the study. Qualitative research requires in-depth analysis and interpretation. Interviews were conducted until data saturation occurred and new insights were no longer being obtained [23]. Because the method used in this study involved interpretation of the findings, it is important to note that other researchers might have other findings and conclusions. However, in this study, all authors discussed the themes and findings throughout the analysis process. This extensive collaboration reduced the possibility of personal interpretations and biased decisions related to the data material [19]. Through the use of a qualitative approach, it was possible to gain deep insight into user perspectives [23], and this provided important insights into, e.g., designing new wearables.

5.1. Implications for practice

Knowing that wearables are associated with relational considerations and that they impact the users' existential experiences, there are pedagogical considerations that are relevant to future studies of PWE using wearables. Healthcare providers should acknowledge the existential impact wearables have on PWE and should verbalize these potential struggles when introducing in-the-wild monitoring.

6. Conclusion

This study contributes novel knowledge of how PWE experience using wearables for in-the-wild monitoring as it brings attention to how it is to experience being placed in the spotlight. For PWE, this

is an existential disturbance of their way of being in the world. In this sense, this study validates the notion of wearables as more than simply technical devices; they also have a significant existential impact. Nevertheless, wearables validate essential aspects of the users' epilepsy, making it easier to accept and understand symptoms related to their diagnosis and reducing uncertainty.

Competing interests

The authors declare no conflicts of interest.

Funding

This study was supported by the NorDigHealth project (087-1.1-18), a EU-Interreg 5A program, and by the Department of Neurology, Zealand University Hospital, Region Zealand, Denmark.

Author JMN also received funds from the Innovation Fund Denmark, Denmark (6152-00005B) and from the Health Research Foundation in Region Zealand, Denmark (A297).

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

We acknowledge all participants who shared their experiences.

Author contributions

JMN and LSO were equal contributors as first authors of this publication. JMN recruited the participants. LSO collected data. CS worked as extern facilitator and academic resource. JMN, TWK, CS, MB and LSO all contributed to the manuscript preparation.

References

- [1] Lu L, Zhang J, Xie Y, Gao F, Xu S, Wu X, et al. Wearable health devices in health care: narrative systematic review. *JMIR MHealth UHealth* 2020;8:e18907. <https://doi.org/10.2196/18907>.
- [2] Lomborg S, Langstrup H, Andersen TO. Interpretation as luxury: heart patients living with data doubt, hope, and anxiety. *Big Data Soc* 2020. <https://doi.org/10.1177/2053951720924436>.
- [3] Rugg-Gunn F. The role of devices in managing risk. *Epilepsy Behav* 2020;103:106456. <https://doi.org/10.1016/j.yebeh.2019.106456>.
- [4] Bruno E, Viana PF, Sperling MR, Richardson MP. Seizure detection at home: Do devices on the market match the needs of people living with epilepsy and their caregivers? *Epilepsia* 2020;61:S11–24. <https://doi.org/10.1111/epi.16521>.
- [5] Hoppe C, Poepel A, Elger CE. Epilepsy: Accuracy of patient seizure counts. *Arch Neurol* 2007;64:1595–9. <https://doi.org/10.1001/archneur.64.11.1595>.
- [6] Karoly P, Goldenholz DM, Cook M. Are the days of counting seizures numbered? *Curr Opin Neurol* 2018;31:162–8. <https://doi.org/10.1097/WCO.0000000000000533>.
- [7] Schmidt D, Schachter SC. Drug treatment of epilepsy in adults. *BMJ* 2014;348:2546. <https://doi.org/10.1136/bmj.g254>.
- [8] Beghi E. Social functions and socioeconomic vulnerability in epilepsy. *Epilepsy Behav* 2019;100. <https://doi.org/10.1016/j.yebeh.2019.05.051>.
- [9] Roberts JI, Sauro K, Jetté N, Osioy K, Knox J, Wiebe S, et al. Using a standardized assessment tool to measure patient experience on a seizure monitoring unit compared to a general neurology unit 2012. 10.1016/j.yebeh.2012.03.002.
- [10] Bruno E, Simblett S, Lang A, Biondi A, Odoi C, Schulze-Bonhage A, et al. Wearable technology in epilepsy: the views of patients, caregivers, and healthcare professionals 2018. 10.1016/j.yebeh.2018.05.044.
- [11] Hoppe C, Feldmann M, Blachut B, Surges R, Elger CE, Helmstaedter C. Novel techniques for automated seizure registration: 'Patients' wants and needs 2015. 10.1016/j.yebeh.2015.08.006.
- [12] Simblett SK, Biondi A, Bruno E, Ballard D, Stoneman A, Lees S, et al. 'Patients' experience of wearing multimodal sensor devices intended to detect epileptic

- seizures: a qualitative analysis. *Epilepsy Behav* 2020;102:106717. <https://doi.org/10.1016/j.yebeh.2019.106717>.
- [13] Beck M, Simony C, Zibrandtsen I, Kjær T. Patient's readiness to body-worn wearables in everyday life: a qualitative study. *Epilepsy Behav* 2020. <https://doi.org/10.1016/j.yebeh.2020.107390>.
- [14] van Manen M. *Phenomenology of Practice: meaning and method in phenomenological research and writing*. 2014.
- [15] Malterud K. *Kvalitative forskningsmetoder for medisin og helsefag*. 4. utgave. Oslo: Universitetsforlaget; 2017.
- [16] van Manen M. *Researching lived experience: human science for an action sensitive pedagogy*. 2. ed. London, Ont: Althouse Press; 1997.
- [17] Kvale S, Brinkmann S. *InterView: Det kvalitative forskningsinterview som håndværk*. 2015.
- [18] Creswell JW. *Qualitative inquiry and research design: choosing among five approaches*. 4. ed. Internatio: SAGE Publications; 2018.
- [19] Polit D, Beck C. *Essentials of nursing research: appraising evidence for nursing practice*. 2018.
- [20] WMA. WMA Declaration of Helsinki – Ethical Principles for Medical Research Involving Human Subjects – WMA – The World Medical Association. World Med Assoc 2018. <https://www.wma.net/policies-post/wma-declaration-of-helsinki-ethical-principles-for-medical-research-involving-human-subjects/> (accessed July 30, 2020).
- [21] Leder D. *The absent body*. Chicago: University of Chicago Press; 1990.
- [22] Mishel MH, Clayton MF. Theories of uncertainty in illness. In: Smith MJ, Liehr PR, editors. *Middle range theory nurs.*, vol. 2. 2nd ed., New York: Springer Publishing Company; 2008, p. 55–84.
- [23] Brinkmann S, Tanggaard L. *Kvalitative metoder: en grundbog*. 3. udgave Kbh: Hans Reitzel; 2020.